

JECRC UNIVERSITY, JAIPUR
School of Engineering
September 2025
B.Tech - 1st Year, Semester-I
Subject Name – Computer Programming and Logical Thinking

Assignment #1

SECTION A (2*5=10 Marks)

1. Why Charles Babbage is called father of computer? mention his contribution.
2. Write five characteristics of Computer?
3. What is cyclic code?
4. Compare Dynamic RAM with Static RAM.
5. What is the Role of 2's Complement in Binary Subtraction?

SECTION B (7*3=21)

1. What is memory hierarchy? Explain the working of Control Unit by a suitable diagram.(3+4)
2. Convert following from one base to another base: (2+2+3)
 - (a) $(AF9)_{16} = (?)_2$
 - (b) $(701)_8 = (?)_2$
 - (c) $(101111.100)_2 = (?)_{10}$
3. (a) Explain importance of ASCII code with example? (3)
(b) Explain primary and secondary memory with suitable examples? (4)

SECTION C (11*3=33 Mark)

1. (a) Explain major developments in 5th generations of computers. (3+4)
(b) Represent -16 in 32-bit floating-point binary format, Mention the role of bias. (4)
2. (a) Explain five generations of computer. Discuss the roll of microprocessor. (5)
(b) Convert the following codes into its equivalent decimal code : (2+2+2)
 - (i) $(11111.101)_2$
 - (ii) $(ABC)_{16}$
 - (iii) $(70067)_8$
3. (a) Explain input and output devices of computer with example? (5)
(b) Evaluate the following: (2+2+2)
 - (i) $10000001-1111 = (?)_2$
 - (ii) $101010 \times 11 = (?)_2$
 - (iii) $10100010+101 = (?)_2$